Lecture 28

Erik An

<2025-11-24 Mon>

Contents

1 Definition of Polynomials **1**
1.1 Notation 2
1.2 Classification by Degree 2
1.3 Roots / Zeros 2

2 Operations **2**

3 Division with Remainder **3**
3.1 Theorem (Polynomial Division) 3
3.2 Example Calculation 3
3.3 Proof of Uniqueness 3

4 Divisibility and Roots **4**
4.1 Definition 4
4.2 Corollary 4
4.2.1 Proof 4
4.3 Theorem: Bound on Roots 4
4.3.1 Proof (by Induction) 4

5 Quadratic Polynomials **4**
5.1 Derivation 5
5.2 Vieta's Formulas 5
5.3 Newton Sums Problem 5

1 Definition of Polynomials

A function f can be represented as:

$$f(x) = a_n x^n + \cdots + a_1 x + a_0$$

where:

- $a_n, \ldots, a_0$ are **coefficients**.
- a_n is the **leading coefficient** (if $a_n \neq 0$).
- $a_n x^n$ is the **leading term**.
- $a_k x^k$ is a **monomial**.

1.1 Notation

- $\mathbb{R}[x]$ is the set of all polynomials with real coefficients.
- $\mathbb{Z}[x]$ is the set of all polynomials with integer coefficients.
- $\deg f = n$ is the **degree** of the polynomial f.

1.2 Classification by Degree

- Degree 1: Linear (Affine)
- Degree 2: Quadratic
- Degree 3: Cubic
- Degree 4: Quartic
- Degree 5: Quintic

1.3 Roots / Zeros

Let $f \in \mathbb{R}[x]$. A number $\alpha \in \mathbb{R}$ is a **root** (or **zero**) of f if $f(\alpha) = 0$.

2 Operations

Let $f(x) = a_n x^n + \cdots + a_0$ and $g(x) = b_m x^m + \cdots + b_0$.

1. **Addition**: $(f + g)(x)$.
2. **Multiplication by a constant**: $(cf)(x) = c \cdot a_n x^n + c \cdot a_{n-1} x^{n-1} + \ldots$.
3. **Product**: $(f \cdot g)(x)$.
4. **Composition**: $f(g(x))$.

3 Division with Remainder

3.1 Theorem (Polynomial Division)

For $a(x), b(x) \in \mathbb{R}[x]$ (with $b(x) \neq 0$), there exist unique polynomials $q(x)$ (quotient) and $r(x)$ (remainder) in $\mathbb{R}[x]$ such that:

$$a(x) = b(x) \cdot q(x) + r(x)$$

where $\deg r < \deg b$. This is analogous to integer division: $a = bq + r$ with $0 \leq r < b$.

3.2 Example Calculation

Divide $a(x) = 2x^4 + 3x^3 - 5x^2 - 7x + 1$ by $b(x) = x^2 - 4x + 2$.

$$\begin{array}{r}
2x^2 + 11x + 35 \\
x^2 - 4x + 2 \overline{\big) 2x^4 + 3x^3 - 5x^2 - 7x + 1} \\
\underline{-(2x^4 - 8x^3 + 4x^2)} \\
11x^3 - 9x^2 - 7x \\
\underline{-(11x^3 - 44x^2 + 22x)} \\
35x^2 - 29x + 1 \\
\underline{-(35x^2 - 140x + 70)} \\
111x - 69
\end{array}$$

Result: $q(x) = 2x^2 + 11x + 35$ and $r(x) = 111x - 69$.

3.3 Proof of Uniqueness

Suppose there are two representations:

$$a(x) = b(x)q_1(x) + r_1(x)$$

$$a(x) = b(x)q_2(x) + r_2(x)$$

Subtracting gives:

$$0 = b(x)(q_1(x) - q_2(x)) + (r_1(x) - r_2(x))$$

$$r_2(x) - r_1(x) = b(x)(q_1(x) - q_2(x))$$

If $q_1 \neq q_2$, then $\deg(RHS) \geq \deg b$. However, $\deg(r_2 - r_1) < \deg b$. This is a contradiction, so $q_1 = q_2$ and $r_1 = r_2$.

4 Divisibility and Roots

4.1 Definition

$a(x)$ is divisible by $b(x)$ (written $b(x)|a(x)$) if there exists $q(x) \in \mathbb{R}[x]$ such that $a(x) = b(x)q(x)$.

4.2 Corollary

Let $f(x) \in \mathbb{R}[x]$. $(x - \alpha)$ divides $f(x)$ if and only if α is a root of f.

4.2.1 Proof

1. **Direction ($\implies$)**: If $(x - \alpha)|f(x)$, then $f(x) = (x - \alpha)q(x)$. Evaluating at α: $f(\alpha) = (\alpha - \alpha)q(\alpha) = 0$.

2. **Direction ($\impliedby$)**: Suppose α is a root ($f(\alpha) = 0$). By division algorithm: $f(x) = (x - \alpha)g(x) + r(x)$, where $\deg r < \deg(x - \alpha) = 1$. Thus, $r(x)$ is a constant r. $f(\alpha) = 0 \cdot g(\alpha) + r \implies 0 = r$. So $f(x) = (x - \alpha)g(x)$, meaning $(x - \alpha)|f(x)$.

4.3 Theorem: Bound on Roots

A non-constant polynomial $f(x) \in \mathbb{R}[x]$ of degree n has at most n real roots.

4.3.1 Proof (by Induction)

- **Base case ($n = 1$)**: Obvious (linear equation has 1 root).

- **Inductive step**: Assume true for degree n. Consider f with $\deg f = n + 1$. If f has a root α, then $f(x) = (x - \alpha)g(x)$, where $\deg g = n$. By hypothesis, g has at most n roots. Therefore, f has at most $n + 1$ roots.

5 Quadratic Polynomials

Given $f(x) = ax^2 + bx + c$. Roots are given by:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

5.1 Derivation

$$ax^2 + bx + c = 0 \iff 4a^2x^2 + 4abx + 4ac = 0$$
$$(2ax)^2 + 2(2ax)b + b^2 - b^2 + 4ac = 0$$
$$(2ax + b)^2 = b^2 - 4ac$$

Let discriminant $D = b^2 - 4ac$.

- If $D > 0$: Two distinct real roots.
- If $D = 0$: One real root ($x_1 = x_2$).
- If $D < 0$: No real roots.

5.2 Vieta's Formulas

If x_1, x_2 are roots:

$$f(x) = a(x - x_1)(x - x_2) = a(x^2 - (x_1 + x_2)x + x_1x_2)$$

Comparing coefficients:

$$\begin{cases} x_1 + x_2 = -\frac{b}{a} \\ x_1x_2 = \frac{c}{a} \end{cases}$$

5.3 Newton Sums Problem

Let $x_1, x_2 \in \mathbb{R}$ be roots of $ax^2 + bx + c$. Let $S_n = x_1^n + x_2^n$. **Prove:** $aS_n + bS_{n-1} + cS_{n-2} = 0$ for $n \geq 2$.